

Lab 01: Classes and Objects

Scenario 1: Library Management System

Raw Data:

- A book titled "Java Programming" written by "John Smith". ISBN: 123-456, Price: 450 Baht. Status: Available.
- A book titled "OOP Concepts" written by "Will Jones". ISBN: 999-888, Price: 590 Baht. Status: Borrowed.

Instructions:

1. **Design a class named Book.**
2. **Identify appropriate Attributes and Methods.**
3. **Implement the following Methods:**
 - `checkOut()`: To change the book's status from "Available" to "Borrowed".
 - `displayDetails()`: To print all information about the book to the console.
4. **Instantiate (create) two objects based on the provided raw data.**
5. **Challenge:** If we want to store the "Return Due Date," should it be a Class Attribute or calculated within a Method? How would you prevent a book that is already "Borrowed" from being checked out again in your code?
6. **Submit the upgrade code and sample output**

Main class

```
package com.library.lab01;

public class LibraryManagementApp {
    public static void main(String[] args) {
        System.out.println("\n" + "=".repeat(60));
        System.out.println("  LIBRARY MANAGEMENT SYSTEM - DEMONSTRATION");
        System.out.println("=".repeat(60));

        // Instantiate two book objects based on the provided raw data
        Book book1 = new Book("Java Programming", "John Smith", "123-456", 450.0,
"Available");
        Book book2 = new Book("OOP Concepts", "Will Jones", "999-888", 590.0,
"Available");

        // Display initial details of both books
        System.out.println("\n--- Initial Book Information ---");
        book1.displayDetails();
        book2.displayDetails();

        // Demonstrate checkOut() method
        System.out.println("\n--- Testing checkOut() Method ---");
        System.out.println("\nAttempting to checkout Book 1 and Book 2 (Available):");
        book1.checkOut();
        book2.checkOut();

        System.out.println("\nAttempting to checkout Book 2 again (Already Borrowed):");
        book2.checkOut();

        // Challenge: Attempt to checkout a book that is already borrowed
        // Demonstrate returnBook() method
        System.out.println("\n--- Testing returnBook() Method ---");
        System.out.println("\nReturning Book 1:");
        book1.returnBook();

        // Display final details
        System.out.println("\n--- Final Book Information ---");
        book1.displayDetails();
        book2.displayDetails();
    }
}
```

Example output:

```
=====
LIBRARY MANAGEMENT SYSTEM - DEMONSTRATION
=====

--- Initial Book Information ---
- Title:           Java Programming
- Author:          John Smith
- ISBN:            123-456
- Price:           450.0 Baht
- Status:          Available
- Return Due Date: N/A (Book is available)

- Title:           OOP Concepts
- Author:          Will Jones
- ISBN:            999-888
- Price:           590.0 Baht
- Status:          Available
- Return Due Date: N/A (Book is available)

--- Testing checkOut() Method ---

Attempting to checkout Book 1 and Book 2 (Available):
  Book 'Java Programming' has been checked out successfully.
  Return Due Date: 2026-02-03
  Book 'OOP Concepts' has been checked out successfully.
  Return Due Date: 2026-02-03

Attempting to checkout Book 2 again (Already Borrowed):
  Error: Book 'OOP Concepts' is already borrowed and cannot be checked out again.

--- Testing returnBook() Method ---

Returning Book 1:
  Book 'Java Programming' has been returned successfully.

--- Final Book Information ---
- Title:           Java Programming
- Author:          John Smith
- ISBN:            123-456
- Price:           450.0 Baht
- Status:          Available
- Return Due Date: N/A (Book is available)

- Title:           OOP Concepts
- Author:          Will Jones
- ISBN:            999-888
- Price:           590.0 Baht
- Status:          Borrowed
- Return Due Date: 2026-02-03
```

Scenario 2: RPG Game Character

Raw Data:

- A warrior named "Arthur", Level 10, Health Points (HP) 1500, Weapon: "Sword".
- A mage named "Merlin", Level 12, Health Points (HP) 800, Weapon: "Staff".

Instructions:

1. **Design a class named Character.**
2. **Create objects named warrior and mage.**
3. **Implement a method attack() that prints a message** (e.g., "Arthur attacks with Sword!") **and a method levelUp().**
4. **Challenge:**
 - If you want to add a "Healer" character who has a unique heal() ability, would you modify the existing Character class or create a new one?
 - Is it necessary to create a separate Weapon class, and what attributes should it have?
5. **Submit the upgrade code and sample output**

Main class

```
package com.rpg.lab01;

public class RPGGameApp {
    public static void main(String[] args) {
        System.out.println("\n" + "=".repeat(70));
        System.out.println("  SCENARIO 2: RPG GAME CHARACTER - DEMONSTRATION");
        System.out.println("=".repeat(70));

        // ===== CREATE WEAPONS =====
        System.out.println("\n[STEP 1] Creating Weapons...");
        Weapon swordWeapon = new Weapon("Sword", "Melee", 25, "Slash Attack");
        Weapon staffWeapon = new Weapon("Staff", "Magic", 20, "Spell Cast");
        Weapon staffHealer = new Weapon("Healing Staff", "Magic", 15, "Healing");

        // ===== CREATE CHARACTERS =====
        System.out.println("\n[STEP 2] Creating Characters...");
        Character warrior = new Character("Arthur", 10, 1500, swordWeapon, "Warrior");
        Character mage = new Character("Merlin", 12, 800, staffWeapon, "Mage");
        Healer healer = new Healer("Elara", 9, 1000, staffHealer, 30);

        // ===== DISPLAY INITIAL STATUS =====
        System.out.println("\n[STEP 3] Initial Character Status:");
        warrior.displayCharacterDetails();
        mage.displayCharacterDetails();
        healer.displayCharacterDetails();

        // ===== TEST ATTACK METHOD =====
        System.out.println("\n[STEP 4] Testing attack() Method:");
        System.out.println("\n--- Warrior's Turn ---");
        int warriorDamage = warrior.attack();

        System.out.println("\n--- Mage's Turn ---");
        int mageDamage = mage.attack();

        // ===== SIMULATE COMBAT =====
        System.out.println("\n[STEP 5] Simulating Combat Scenario:");
        System.out.println("\n--- Mage takes damage from Warrior ---");
        mage.takeDamage(warriorDamage);

        System.out.println("\n--- Warrior takes damage from Mage ---");
        warrior.takeDamage(mageDamage);

        // ===== TEST LEVEL UP =====
```

```

System.out.println("\n[STEP 6] Testing levelUp() Method:");
System.out.println("\n--- Arthur levels up! ---");
warrior.levelUp();

System.out.println("\n--- Merlin levels up! ---");
mage.levelUp();

// ===== TEST HEALER ABILITIES =====
System.out.println("\n[STEP 7] Testing Healer Special Abilities:");
System.out.println("\n--- Healer heals themselves ---");
healer.displayCharacterDetails();
healer.heal();

System.out.println("\n--- Healer heals the Warrior ---");
healer.healAlly(warrior);
}
}

```

Example Output:

```

=====
SCENARIO 2: RPG GAME CHARACTER - DEMONSTRATION
=====

[STEP 1] Creating Weapons...
[STEP 2] Creating Characters...
[STEP 3] Initial Character Status:

--- ARTHUR ---
Class: Warrior
Level: 10
Health Points: 1500/1500
Weapon: Sword (Type: Melee, Damage: 25, Ability: Slash Attack)

--- MERLIN ---
Class: Mage
Level: 12
Health Points: 800/800
Weapon: Staff (Type: Magic, Damage: 20, Ability: Spell Cast)

--- ELARA ---
Class: Healer
Level: 9
Health Points: 1000/1000
Weapon: Healing Staff (Type: Magic, Damage: 15, Ability: Healing)
Healing Power: 57

```

```
[STEP 4] Testing attack() Method:

--- Warrior's Turn ---
Arthur (Warrior) attacks with Sword!
    🗡️ Attack Damage: 45 (Weapon Base: 25 + Level Bonus: 20)

--- Mage's Turn ---
Merlin (Mage) attacks with Staff!
    🪄 Attack Damage: 44 (Weapon Base: 20 + Level Bonus: 24)

[STEP 5] Simulating Combat Scenario:

--- Mage takes damage from Warrior ---
Merlin takes 45 damage! Current HP: 755/800

--- Warrior takes damage from Mage ---
Arthur takes 44 damage! Current HP: 1456/1500

[STEP 6] Testing levelUp() Method:

--- Arthur levels up! ---
Arthur leveled up to Level 11!
    🏠 Max Health increased to 1510 (full heal applied)

--- Merlin levels up! ---
Merlin leveled up to Level 13!
    🏠 Max Health increased to 810 (full heal applied)

[STEP 7] Testing Healer Special Abilities:

--- Healer heals themselves ---

--- ELARA ---
Class: Healer
Level: 9
Health Points: 1000/1000
    Weapon: Healing Staff (Type: Magic, Damage: 15, Ability: Healing)
    Healing Power: 57
Elara (Healer) casts a healing spell!
    ✨ Healing Amount: 57 (Base: 30 + Level Bonus: 27)
    ❤️ Restored 57 HP! Current HP: 1000/1080

--- Healer heals the Warrior ---
Elara (Healer) casts healing spell on Arthur!
    ✨ Healing Amount: 57
    ❤️ Restored 57 HP to Arthur! Their HP: 1510
```